

ADITYA CHATURVEDI

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Education

Jaypee Institute Of Information Technology

July 2023 – July 2027

Bachelor of Technology in Computer Science and Engineering | CGPA: 7.5/10.0

Noida, India

Seth Anandram Jaipuria School

2022 – 2023

Senior Secondary (Class XII), CBSE | Percentage: 90%

Ghaziabad, India

Seth Anandram Jaipuria School

2020 – 2021

Secondary (Class X), CBSE | Percentage: 95%

Ghaziabad, India

Technical Skills

Languages: C++, Python, Java, Kotlin, Dart (Flutter), SQL

CS Fundamentals: Data Structures, Algorithms, OOP, Operating Systems, DBMS, Computer Networks, Distributed Systems

Mobile & Backend: Android SDK, Jetpack Compose, Firebase, REST APIs, Redis, FAISS, Room DB, Retrofit, FastAPI

AI/ML & Research: PyTorch, PyTorch Geometric, Graph Neural Networks, Diffusion Models, RAG, LLMs, LangChain

Developer Tools: Git/GitHub, Docker, Streamlit, Postman, MySQL, Figma

Experience

GirlScript Summer of Code (GSSoC'25)

July 2025 – October 2025

Open-Source Contributor

Remote

- Engineered **security enhancements** for **Flutter authentication** via **OTP expiry** and **Twilio API** integration, patching critical vulnerabilities and reducing **unauthorized access by 40%**.
- Redesigned **ShopSmart** profile UI using **Material Design** and **Lottie animations**, boosting **user engagement by 25%** and increasing average **session duration by 15%**.
- Merged **15+ pull requests** across **10+ repositories**, resolving performance bottlenecks and security vulnerabilities; changes reviewed and approved by project maintainers.

Projects

ContextCore (RAG Document Intelligence) | Python, FastAPI, FAISS, Redis | GitHub | Live Demo

Jan 2026 – Mar 2026

- Built a **hybrid retrieval pipeline** combining **FAISS vector search** and **BM25 keyword matching** fused via **Reciprocal Rank Fusion**, outperforming vector-only recall; engineered **SSE-based token streaming** for sub-second real-time responses.
- Reduced repeated query latency from **2.5s to 4ms** via **Redis caching** with TTL-based invalidation, eliminating redundant LLM calls; enforced **per-user document isolation** using **JWT auth** with isolated FAISS vectorstores.
- Designed a **stateless, horizontally scalable backend** with **multi-turn conversational memory** via history-injected prompts and full session persistence (save/load/delete) across concurrent users.

RA-MDM (Rhythm-Aware Motion Diffusion) | Python, PyTorch, Diffusion Models | GitHub

Nov 2025 – Feb 2026

- Engineered a **rhythm-conditioned diffusion framework** achieving **0.94 beat alignment** (6.5% improvement over HMDM); designed custom **Librosa-based audio feature extraction** pipeline with exponential averaging for rhythm synchronization across 4 dance styles.
- Extended **DDPM** with **style-adaptive FiLM layers** and learned noise schedules for **multimodal audio-to-motion synthesis** on AIST++ dataset; achieved **18% lower FID** and **4.7/5 user perceptual rating** vs RNN/VAE/GAN/HMDM baselines.
- Published on **ResearchGate**; demonstrates research rigor with comprehensive ablation studies, quantitative validation, and comparative baseline analysis.

GAT Land Cover Classification | Python, PyTorch, PyTorch Geometric | GitHub

Nov 2025 – May 2026

- Engineered an end-to-end **Graph Attention Network** pipeline on **EuroSAT** (27,000 images, 13 spectral bands), reducing image complexity **50×** via **SLIC superpixel graphs** with **21-dimensional node features**.
- Achieved **76.9% accuracy** and **100% F1 on water body detection** using physics-grounded NDVI thresholding, outperforming CNN and Random Forest baselines by up to **14.2%**.
- Implemented a **2-layer GAT with 8 attention heads** to capture irregular spatial boundaries; submitted research paper to **IC3-2026 (IEEE, SCOPUS-indexed)** as the first reported **SLIC+NDVI+GAT** application on EuroSAT benchmark.

Achievements

- Competitive Programming:** Rated **1720+** on [LeetCode](#) (**Top 5%**); solved **1000+** problems across LeetCode, GFG, and Coding Ninjas.
- Research:** Authored **2 peer-reviewed papers** — “*Rhythm-Aware Motion Diffusion Model*” (**Published on ResearchGate**) and “*GAT for Land Cover Classification*” (**IC3-2026, IEEE SCOPUS-indexed**).
- Open Source:** **GSSoC 2025** — engineered security enhancements reducing unauthorized access by 40%; merged **15+ PRs** across **10+ repositories** with code review approval.
- Simulations:** **JPMorgan Chase** (Kafka, Spring Boot, REST APIs) and **Deloitte** (Tech Dashboards, Data Analysis).
- Community:** Organized campus-wide coding contests as **Volunteer** at Knuth Programming Hub, IIIT.